



Turning in Code

Quick facilitation notes & answer key

Goal

Students read a for-loop with a move AND a turn inside, and see that repeating move+turn draws a shape. Turn angles are given (no deriving). Sets up drawing a full square next sheet.

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

Python Turtle projected (or a web playground); the printed worksheet + a pencil per student.

Run it (about 30 min)

1. Recall last sheet's forward-only loop; add a `right(90)` inside and run it.
2. Trace the `range(3)` path together — 3 sides, 3 turns, open.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; predict that `range(4)` closes the square (next sheet).

Answer key (checked by the run-gate)

Ex 1 — 3 forwards; 3 turns.

Ex 2 — a square (`range(4)` adds the 4th side and closes it).

Ex 3 — `range(4)`, `bit.right(90)`.

Ex 4 (challenge) — 360° in total ($4 \times 90 =$ a full turn).

Watch for & differentiate

Common slip: forgetting the turn also repeats — both lines inside run each time.

Support: have students trace with a finger, saying 'forward, turn' each repeat.

Extend: ask the total turning for `range(3)` at right 90 (270°).

Success looks like

The student counts the moves and turns (Ex 1), predicts `range(4)` closes a square (Ex 2), completes the loop (Ex 3), and adds the total turning (Ex 4).